

LAB RESOURCE UTILIZATION PLATFORM

Document 6

System Architecture

Infosys Springboard Internship

1. Introduction

The Lab Resource Utilization Platform follows a layered architecture with a React-based client, Spring Boot backend services, secure API gateway, messaging layer, and PostgreSQL database. The architecture is designed to provide secure, scalable, and efficient laboratory resource management.

2. Overall System Architecture

DOCUMENT 6 – SYSTEM ARCHITECTURE

1. INTRODUCTION

The Lab Resource Utilization Platform is a web-based system designed to streamline the management, booking, monitoring, and maintenance of laboratory resources. The system follows a modular monolithic architecture using Spring Boot, with React.js as the frontend. It integrates multiple external services and uses PostgreSQL for primary data storage, Redis for caching, and Apache Kafka / RabbitMQ for messaging and real-time communication.

2. OVERALL SYSTEM ARCHITECTURE

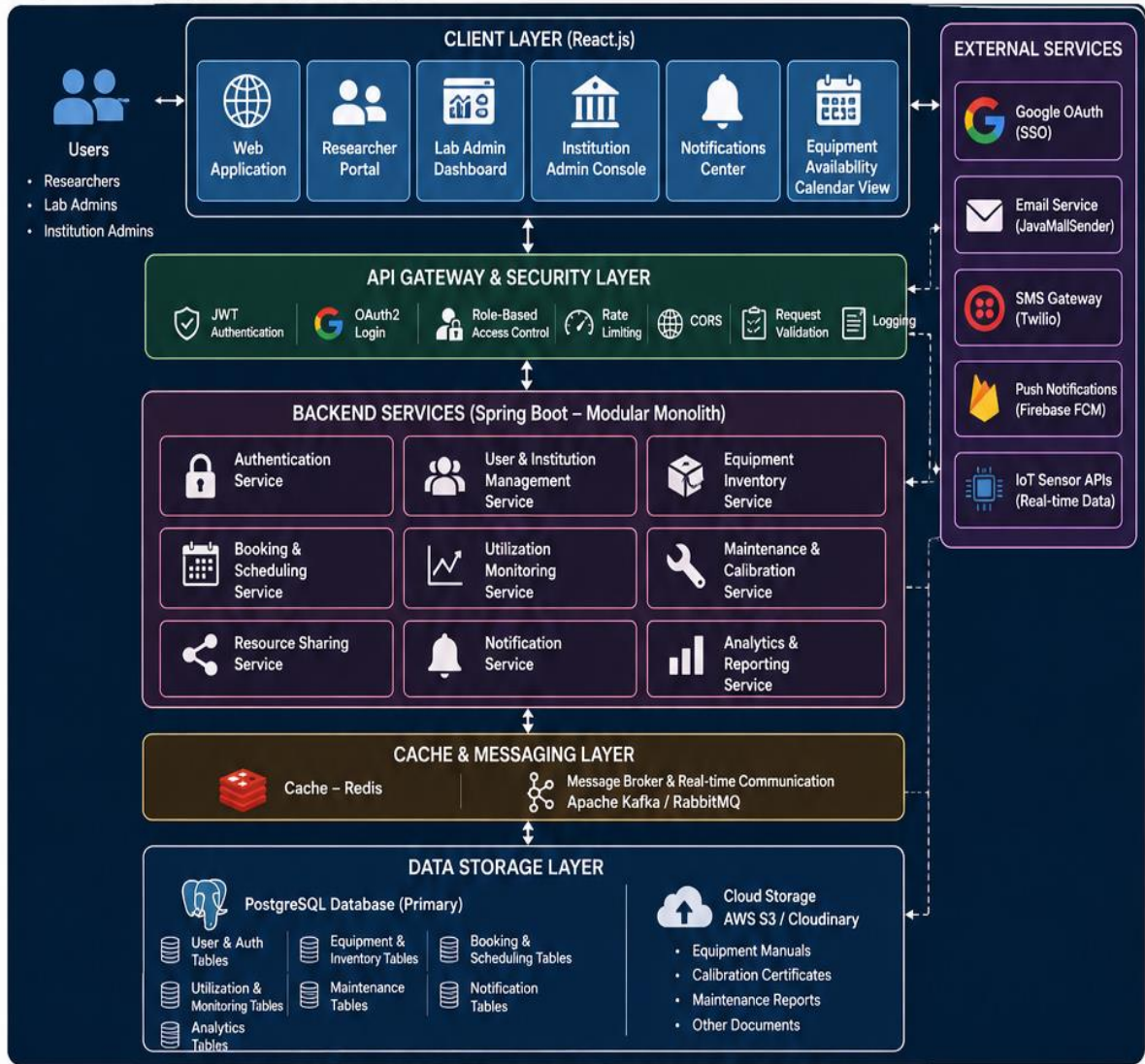


Figure 6.1: Overall System Architecture of Lab Resource Utilization Platform

The architecture follows a layered approach ensuring separation of concerns, scalability, security, and high availability. The frontend communicates with the backend through a secure API Gateway. Modular services handle specific business functionalities, while external services and messaging systems enable integration and real-time capabilities.

3. ARCHITECTURE COMPONENTS

Layer / Component	Description
 Client Layer	React.js based web application providing different role-based interfaces for researchers, lab admins and institution admins.
 API Gateway & Security Layer	Handles authentication, authorization, request validation, rate limiting, CORS and logging for secure and controlled access.
 Backend Services	Modular Spring Boot services implementing core business functionalities.
 External Services	Third-party services for authentication, email, SMS, push notifications and IoT data.
 Cache & Messaging Layer	Redis for caching frequent data; Kafka/RabbitMQ for asynchronous messaging and real-time communication.
Data Storage Layer	PostgreSQL for structured data storage and AWS S3/Cloudinary for document storage.

